THE FERTILIZERS — FULL MIND MAP

Chemistry · Chapter 10 · Topic-wise · Fully Explained · Hinglish · SSC/PCS/UPSC Ready

Topics in this chapter:

1. Plant Nutrients, NPK & Fertilizer Basics

- Fertilizer wo natural ya chemical substance hai jo soil fertility, plant growth aur yield improve karne ke liye add kiya jata hai. N, P aur K major/macronutrients hain: Nitrogen growth-development ke liye, Phosphorus healthy root development ke liye aur Potassium stomata ke opening-closing regulation me important hai. [Q1–Q4]
- Source ke hisaab se 1 metric ton organic manure se roughly 1.5–3 kg phosphorus mil sakta hai, isliye phosphorus requirement ko meet karne me chemical fertilizer important role play karta hai. Wheat cultivation me Nitrogen fertilizer especially important maana gaya hai. [Q1–Q2]
- Plants ko Carbon, Hydrogen aur Oxygen mainly CO_2 aur water se milte hain, jabki N, P, K, Ca, Mg, B, Cl, Cu, Fe, Mn, Mo, Zn jaise nutrients soil/inorganic salts se mil sakte hain. Isliye listed options me Hydrogen fertilizer se directly supplied element nahi maana gaya hai. [Q3]

⚠ EXAM TRAP — Q4 — Source itself marks the question disputed

- ▸ Nitrogen → growth/development, Phosphorus → root development, Potassium → stomatal regulation — ye matches correct hain.
- ▸ Boron micronutrient hai; source disease-resistance role bhi accept karta hai. Isliye given options me koi clearly incorrect pair nahi bachta, so source answer = * (disputed).

2. Urea: Composition, Solubility & Nitrogen Content

- Urea/Carbamide ka formula $CO(NH_2)_2$ hai. Ye nitrogen-containing organic compound hai aur isme Carbon, Hydrogen, Nitrogen aur Oxygen chaaron elements present hote hain. Nitrogen do amide (NH_2) groups me hota hai, isliye urea me nitrogen ka form = amide form. [Q6,Q8–Q9]
- Urea me nitrogen approximately 46–47% hota hai. Isi basis par 1 kg nitrogen supply karne ke liye required urea ≈ 100/46 = 2.17 kg, yani roughly 2.2 kg. [Q7,Q11]
- Listed fertilizers me 20°C par urea highly water-soluble hai, isi high solubility aur leaf absorption ke karan urea foliar application/spray ke liye bhi bahut popular fertilizer hai. [Q10,Q19]

3. Nitrogen Fertilizers, Soil Acidity & Environmental Effects

- Ammonium sulphate soil ko comparatively zyada acidify kar sakta hai; acidic soil ka pH 7 se kam hota hai. Nitrogenous fertilizers ka excessive/inappropriate use soil acidity increase kar sakta hai aur unused nitrate groundwater me leach hokar water pollution cause kar sakta hai. [Q5,Q13]
- Excess artificial nitrogen se nitrogen-fixing microorganisms ka proliferation expected nahi hota; source ke explanation ke according natural nitrogen-fixing bacterial activity suppress bhi ho sakti hai. Isliye Q13 me statements 2 aur 3 correct hain. [Q13]

⚠ EXAM TRAP — Nitrogen fertilizer ka “more is better” rule nahi hota

- ▸ Nitrogen essential hai, lekin overuse se soil acidification, nitrate leaching aur environmental pollution ho sakta hai.
- ▸ Exam me fertilizer benefit aur fertilizer overuse ke environmental effects ko alag-alag context me read karo.

4. Urea Production, Neem-coated Urea & Fertilizer Policy Context

- Industrial urea synthetic ammonia (NH_3) aur carbon dioxide (CO_2) se produce hota hai. Ammonia production me hydrogen commonly natural gas feedstock se obtain kiya jata hai. [Q12,Q14]
- Neem-coated urea me neem coating nitrification/urea conversion ko slow karke nitrogen release ko more gradual banati hai. Isse nitrogen-use efficiency improve ho sakti hai aur nutrient loss reduce karne me help milti hai. [Q15]
- Source-era policy question me statement 1 incorrect maana gaya because urea MRP Government-controlled context me tha; statements 2–3 correct: ammonia natural-gas route se aur elemental sulphur petroleum-refinery by-product route se fertilizer industry me linked hai. [Q12]

⚠ EXAM TRAP — Q12 is policy/time-context sensitive

- ▸ Question wording “At present” source-exam period ke context me hai. Is mind map me source explanation preserve ki gayi hai.
- ▸ Static chemistry fact: urea production uses NH_3 + CO_2; sulphur/phosphoric-acid fertilizer chain refinery by-product sulphur se linked ho sakti hai.

5. DAP, NPK, Foliar Feeding, Humus & Green Manure

- DAP = Diammonium Phosphate, formula $(NH_4)_2HPO_4$. Standard fertilizer grade ko commonly 18-46-0 ke form me express kiya jata hai: about 18% Nitrogen aur 46% P_2O_5. [Q16]
- NPK mixed/complex fertilizer hai jo Nitrogen + Phosphorus + Potassium combine karta hai. Urea, Super Phosphate aur Potassium Nitrate common chemical-fertilizer examples hain; listed options me Sodium Sulphate ko chemical fertilizer nahi maana gaya. [Q17–Q18]
- Humus decomposed plant-animal matter se banne wala dark organic material hai aur soil me organic colloid ki tarah behave karta hai. Green manuring me generally leguminous crops grow karke soil me incorporate kiye jate hain; listed options me Sunhemp (sanai) best green-manure crop hai. [Q20–Q21]

✓ Source-question mapping preserved: Q1–Q21